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10 February 1968
SUPERSEDING
MIL-C-22931A
27 March 1964

MILITARY SPECIFICATION

CABLES, RADIO FREQUENCY, SEMIRIGID,

COAXIAL, SEMI-AIR-DIELECTRIC,

GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers semi-air-dielectric, coaxial, semirigid radio frequency cables with smooth, corrugated, or braided outer conductors with outside diameters ranging from .500 to 3.125 inches, with an impedance of 50 or 75 ohms (see 3.1). The operating temperature ranges are -55°C to +80°C or -55°C to +200°C, as specified (see 3.1), and the storage temperature capability is -65°C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic Molding Material, Polyethylene, Low and Medium Density.
QQ-C-576 - Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet, and Strip).
QQ-W-343 - Wire, Electrical and Nonelectrical, Copper, (Uninsulated).

FSC 6145

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MIL-P-116 - Preservation, Methods of.
MIL-R-5757/21 - Relays, Electromagnetic, Hermetically Sealed, Latching DPDT Contacts, Low Level to 2 Amperes.
MIL-C-12000 - Cable, Cord, and Wire, Electric, Packaging and Packing of.
MIL-C-45662 - Calibration System Requirements.

(See supplement 1 for list of applicable specification sheets.)

~~STANDARDS~~

FEDERAL

Fed. Test Method Std. No. 228 - Cable and Wire, Insulated; Method of Testing.
Fed. Test Method Std. No. 406 - Plastics, Methods of Testing.

MILITARY

MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronics and Electrical Component Parts.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-456 - Electronic Parts, Date and Source Coding for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

B188-66 - Standard Specification for Seamless Copper Bus Pipe and Tube.
B209-66 - Standard Specification for Aluminum-Alloy Sheet and Plate.
B210-66 - Standard Specification for Aluminum-Alloy Drawn Seamless Tubes.
B221-66 - Standard Specification for Aluminum-Alloy Extruded Bars, Rods, Shapes, and Tubes.
B234-66 - Standard Specification for Aluminum-Alloy Drawn Seamless Tubes for Condensers and Heat Exchangers.
D150-65T - Tentative Methods of Test for A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.
D648-56 - Standard Method of Test for Deflection Temperature of Plastics Under Load.
D703-54 - Standard Specifications for Polystyrene Molding and Extrusion Materials.
D1248-65T - Tentative Specification for Polyethylene Molding and Extrusion Materials.

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(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheet the latter shall govern.

3.2 Materials. The materials used in the construction of the cable shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the cable to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Inner conductor. The inner conductor shall be solid or tubular as specified (see 3.1) and shall be in accordance with 3.2.1.1, 3.2.1.2, or 3.2.1.3.

3.2.1.1 Solid. Solid conductors shall be wire conforming to type S of QQ-W-343.

3.2.1.2 Tubular. Tubular conductors shall conform to the requirements of ASTM Publication B188-66.

3.2.1.3 Tubular, with seam or weld. Tubular conductors, with seam or weld, shall be manufactured from copper strip that conforms to QQ-C-576.

3.2.1.4 Joints. There shall be no joints in solid conductors made subsequent to the last drawing operation. Joints in the inner conductors shall be brazed or silver soldered using a nonacid flux, so that the diameter shall not be increased and there shall be no lumps or sharp projections.

3.2.2 Dielectrics. The dielectrics used in the construction of cables shall be as specified for each specified application (see 3.1), and shall be in accordance with 3.2.2.1, 3.2.2.2 or 3.2.2.3.

3.2.2.1 Polyethylene. Polyethylene shall be in accordance with ASTM Publication D-1248-65T.

3.2.2.2 Polystyrene. Polystyrene tapes shall be made from biaxially oriented sheets of pure polystyrene, free from colorants or lubricants, and in general shall meet the requirements of ASTM Publication D703-64. In addition, the polystyrene shall have a minimum heat distortion temperature of 92°C, measured in accordance with ASTM Publication D648-56; a dielectric constant of 2.5 at 1 x 10^6 hertz (Hz); and a dissipation factor of 0.0003 maximum at 1 x 10^6 Hz, both measured in accordance with ASTM Publication D150-65T.

3.2.2.3 TFE fluorocarbon. TFE fluorocarbon shall be polytetrafluorethylene. The dielectric constant and dissipation factor (tangent of loss angle) shall not exceed 2.1 and 0.0003, respectively, when measured in accordance with method 4021 of FED. Test Method STD. 406.

3.2.3 Outer conductors. The outer conductor (or shield) of the cable shall be braided, smooth, or corrugated metal tube, as specified (see 3.1), and shall be in accordance with 3.2.3.1 to 3.2.3.2.2. The conductor may be formed in any manner, provided the requirements of this specification are met.

3.2.3.1 Braid. The individual strands of the braid shall conform to type S of QQ-W-343.

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3.2.3.2 Tubes and strips.

3.2.3.2.1 Aluminum. Aluminum tubes shall conform to the requirements of ASTM Publications B210-66, B234-66, or B221-66, as applicable. Aluminum strips may be used to form tubes provided they meet the requirements of ASTM Publication B209-66.

3.2.3.2.2 Copper. Copper tubes shall conform to the requirements of ASTM Publication B188-66. Copper strips shall conform to the requirements of QQ-C-576.

3.2.4 Jacket (when applicable). The jacket shall consist of a continuous sheath of polyethylene conforming to type III, grade 8 of L-P-390.

3.2.4.1 Thickness. The minimum wall thickness of the jacket at any point shall be as specified (see 3.1).

3.3 Dimensions. The dimensions of the cables shall be as specified (see 3.1).

3.4 Electrical performance.

3.4.1 Continuity. When cables are tested as specified in 4.6.2, each conductor in each reel of cable shall be continuous.

3.4.2 Voltage withstand. When cables are tested as specified in 4.6.3, the completed cable shall withstand the potential specified (see 3.1) without breakdown.

3.4.3 Insulation defects, spark (when applicable, see 3.1). When cables with jackets are tested as specified in 4.6.4, the jacket shall be continuous without cracks, breaks or holes.

3.4.4 Insulation resistance. When cables are tested as specified in 4.6.5, the insulation resistance per 1,000 feet shall be not less than 100,000 megohms.

3.4.5 Attenuation. When cables are tested as specified in 4.6.6, the attenuation shall be as specified (see 3.1).

3.4.6 Velocity. When cables are tested as specified in 4.6.7, the velocity shall be as specified (see 3.1).

3.4.7 Capacitance. When cables are tested as specified in 4.6.8, the capacitance shall be as specified (see 3.1).

3.4.8 Impedance. When cables are tested as specified in 4.6.9, the impedance shall be as specified (see 3.1).

3.4.9 Voltage standing wave ratio (VSWR). When cables are tested in accordance with 4.6.10, the VSWR of the sample unit shall not exceed the value specified (see 3.1) as the initial VSWR. The connectors shall be included as part of the cable, and shall comply with the requirements specified herein.

3.5 Environmental.

3.5.1 Pressure tightness (when applicable, see 3.1). When tested as specified in 4.6.11, cables with smooth or corrugated outer conductors shall be capable of retaining the internal pressure specified (see 3.1), with a pressure drop of not more than 1 pound per square inch gage (psig) over a period of 24 hours.

3.5.2 Cold bend (when applicable, see 3.1). When specified, jacketed cable shall be tested as specified in 4.6.12, and there shall be no evidence of cracking or splitting of the cable jacket.

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3.5.3 Bending (when applicable, see 3.1). When specified, finished cables shall be tested in accordance with 4.6.13. Following the test, the cable shall meet the requirements of 3.4.9 and 3.5.2. When the alternate procedure is used, there shall be no evidence of kinking, wrinkling or cracking.

3.5.4 Temperature cycling (when applicable, see 3.1). When specified, cables shall be tested as specified in 4.6.14. There shall be no evidence of mechanical damage, and the capacitance and VSWR shall be as specified (see 3.1).

3.5.5 Thermal shock (when applicable, see 3.1). When specified, cables shall be tested as specified in 4.6.15. There shall be no flaring of any layer, and unless otherwise specified (see 3.1), the measurements following exposure shall not vary from the original measurement by more than the following:

Core	-	1/8 inch
Insulating layer	-	3/8 inch
Outer conductor	-	1/8 inch
Outer jacket	-	3/4 inch

3.5.6 Vibration, high frequency (when applicable, see 3.1). When specified, cables shall be tested as specified in 4.6.16. There shall be no conductor discontinuity nor evidence of damage.

3.5.7 Shock (when applicable, see 3.1). When specified, cables shall be tested as specified in 4.6.17. There shall be no conductor discontinuity nor evidence of damage.

3.6 Marking. The cables shall be marked in accordance with MIL-STD-130 with the complete part number, the manufacturer's code symbol, and the manufacturer's name, at least every 2 feet. The manufacturer's code symbol shall be in accordance with MIL-STD-456. Marking shall not permanently indent, deform, or otherwise damage the jacket or outer conductor.

3.7 Workmanship. Cables shall be processed in such a manner as to be uniform in quality and shall be free from any burrs, die marks, chatter marks, foreign material, and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the cables, are in accordance with the applicable referenced specification or requirements prior to such fabrication.

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TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Wire and braid -	3.2.1.1, 3.2.3.1	QQ-W-343
Copper - - - - -	3.2.1.3, 3.2.3.2.2	QQ-C-576, ASTM B188-66
Polyethylene - -	3.2.2.1, 3.2.4	ASTM D1248-65T, L-P-390
Polystyrene - -	3.2.2.2	ASTM D150-65T, D648-56, D703-64
TFE fluorocarbon	3.2.2.3	Fed. Test Method Std 406
Aluminum - - - -	3.2.3.2.1	ASTM B209-66, B210-66, B221-66, B234-66

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.5.2.1.5, delivery of products which have passed groups A and B inspections shall not be delayed pending results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all cable covered by a single specification sheet, produced under essentially the same conditions, and offered for inspection at the same time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II, in the order shown.

TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Test paragraph	Sampling plan
Subgroup 1			
Visual and mechanical - - - - - -	3.1, 3.3, 3.6 and 3.7	4.6.1	See paragraph 4.5.1.2.1
Subgroup 2			
Continuity - - - - - - - - - - -	3.4.1	4.6.2	100% inspection required
Voltage withstand - - - - - - -	3.4.2	4.6.3	
Insulation defects, spark (when applicable) - - - - - -	3.4.3	4.6.4	
Insulation resistance - - - - - -	3.4.4	4.6.5	
Pressure tightness (when applicable) - - - - - - - - - -	3.5.1	4.6.11	

4.5.1.2.1 Sampling plan. Statistical sampling and inspection for subgroup 1 shall be in accordance with table III, except that the number of sample units shall be not more than two times the number of reels in the inspection lot. No more than two sample units shall be selected from each reel of cable. When two sample units are required from one reel, they shall be cut from each end of the reel. For subgroup 2, each reel in the inspection lot shall be tested.

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TABLE III. Sampling plan for group A inspection.

Inspection lot size Cable length (feet)	Sample units	Accept number	Reject number
0 to 16,000	5	0	1
16,001 to 26,000	8	0	1
26,001 to 64,000	13	1	2
64,001 to 160,000	20	1	2
160,001 to 440,000	32	2	3
440,001 to 2,200,000	50	3	4
2,200,001 and over	80	5	6

4.5.1.2.2 Sample unit. For subgroup 1, a sample unit is a piece of cable three feet in length and cut from the reel of cable. For subgroup 2, the complete length of cable on each reel shall be tested.

4.5.1.2.3 Defective unit. A defective unit is any sample unit failing any test in any test group. The reel of cable from which the sample unit was taken will be considered defective.

4.5.1.2.4 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.2.5 Disposition of sample units. Sample units, and the reels of cable from which the sample units were taken, which failed one or more of the inspections shall not be delivered on the contract or order. However, insulation defects, spark failures may be repaired or the cable cut at this point (see 4.6.4), and shipped.

4.5.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table IV, in the order shown, and shall be made on reels of cable which have been subjected to and have passed the group A inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with variability unknown-standard deviation method of MIL-STD-414, at level IV for AQL 4.0 percent defective.

TABLE IV. Group B inspection.

Test	Requirement paragraph	Test paragraph
Attenuation 1/ -	3.4.5	4.6.6
Velocity - - - -	3.4.6	4.6.7
Capacitance - -	3.4.7	4.6.8
Impedance - - -	3.4.8	4.6.9
VSWR - - - - - -	3.4.9	4.6.10

1/ At all specified frequencies.

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4.5.1.3.2 Sample unit. A sample unit is a piece of cable cut from the reel of cable and of the length specified for the applicable test method.

4.5.1.3.3 Defective unit. A defective unit is any sample unit failing any test or inspection of table IV. The reel of cable from which the sample unit was taken will be considered defective.

4.5.1.3.4 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.5 Disposition of sample units. Sample units and the reels of cable from which the sample units were taken may be delivered on the contract or order, if the sample units have passed all the group B inspection and the lot is accepted.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C inspection.

4.5.2.1 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table V, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

TABLE V. Group C inspection.

Test	Requirement paragraph	Test paragraph
Cold bend (when applicable) - - -	3.5.2	4.6.12
Bending (when applicable) - - - -	3.5.3	4.6.13
Temperature cycling (when applicable) - - - - - - - - - -	3.5.4	4.6.14
Thermal shock (when applicable) -	3.5.5	4.6.15
Vibration, high frequency (when applicable) - - - - - - - - - -	3.5.6	4.6.16
Shock (when applicable) - - - - -	3.5.7	4.6.17

4.5.2.1.1 Sampling plan. Sample units shall be selected from each 6-months' production of cable covered by a single specification sheet, in accordance with table VI, except that the number of sample units shall be not more than two times the number of reels in the inspection lot. No more than two sample units shall be selected from each reel of cable. When two sample units are required from one reel, they shall be cut from each end of the reel. The sample units shall be selected from different production runs throughout the 6-month period. No failures shall be permitted for group C inspection.

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TABLE VI. Sampling plan for group C inspection.

Lot size (Six Months Production) Cable length	Sample units
(feet)	
0 to 16,000	0
16,001 to 26,000	1
26,001 to 64,000	2
64,001 to 160,000	3
160,001 to 440,000	4
440,001 to 2,200,000	5
2,200,001 and over	8

4.5.2.1.2 Sample unit. A sample unit is a piece of cable cut from the reel of cable of the length specified for the applicable test method.

4.5.2.1.3 Defective unit. A defective unit is any sample unit failing any test or inspection of table V. The reel of cable from which the sample unit was taken will be considered defective.

4.5.2.1.4 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order. The reel from which the sample unit was taken may be shipped, if the sample unit was not defective, and the lot is accepted.

4.5.2.1.5 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action as successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the contracting officer.

4.5.3 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. The cable shall be examined to verify that the design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.1, 3.3, 3.6 and 3.7.) A micrometer caliper or an instrument of equal accuracy shall be used to determine the proper dimensions.

4.6.2 Continuity (see 3.4.1). A direct current (dc) potential of 6 volts maximum shall be applied, through an appropriate indicator, to the inner and outer conductors of the reel of cable. The voltage may be applied to the conductors individually or in series.

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4.6.3 Voltage withstand (see 3.4.2). The cables shall be tested in accordance with method 6111 of FED. TEST METHOD STD. NO. 228 except that the cable shall not be immersed in water but shall be tested dry and the use of direct current (dc) voltage is acceptable. The following details shall apply:

(a) The test shall be performed on completed cable only.
(b) The test voltage shall be as specified (see 3.7).
(c) The potential shall be applied to the inner conductor with the outer conductor grounded.

4.6.4 Insulation defects, spark (when applicable, see 3.4.3). Jacketed cables shall be tested in accordance with method 6211 of FED. TEST METHOD STD. NO. 228. The following details shall apply:

(a) The test voltage shall be a 60 Hz root-mean-square (rms) voltage as specified in table VII.
(b) The potential shall be applied between the outer conductor and the outer surface of the jacket. A puncture of the jacket by the applied voltage shall constitute a point of failure. The cable may be cut at this point, or it may be repaired to the satisfaction of the Government.

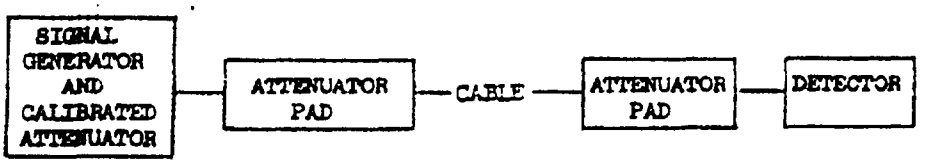
TABLE VII. Voltage for spark test.

Jacket thickness (minimum)	Voltage
(inch)	(rms)
0.035	7,000
.060	9,000
.065	10,000

4.6.5 Insulation resistance (see 3.4.4). The cable shall be tested in accordance method 6031 of FED. TEST METHOD STD. NO. 228, except that the cable shall not be immersed in water but shall be tested dry. The following details shall apply:

(a) The test shall be performed on completed cable only.
(b) The test voltage shall be not less than 200 volts.
(c) The potential shall be applied to the inner conductor with the outer conductor grounded.

4.6.6 Attenuation (see 3.4.5). The attenuation, expressed in decibels (dB) per 100 feet, shall be measured at a sufficiently low-power level that the resulting temperature rise will be negligible. An acceptable method for measuring attenuation is as follows:



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In the block diagram, a suitable length of cable with an attenuation greater than the measuring accuracy of the equipment is inserted between the connectors. The signal generator and calibrated attenuator are adjusted to produce a reasonable indication at the detector, when the detector is tuned. The detector reading is noted, and the calibrated attenuator output level is recorded. The cable under test is then withdrawn and the circuit completed with the connectors (or a very short length of cable). With the detector tuned, the calibrated attenuator is readjusted to reproduce the original reading at the detector and the attenuator output level is again recorded. Attenuation is then computed as follows:

$$A = \frac{100}{L} \text{ (Difference in calibrated attenuator reading in dB)}$$

Where:
A = attenuation in dB per 100 feet.
L = length of cable under test in feet.

For measurement at frequencies of 400 megahertz (MHz), or less, the characteristic impedance of the attenuator pads and connectors shall preferably be the same as that of the cable under test. For measurement at frequencies of 1,000 MHz or above, the attenuator pads, connectors, and test cable shall be matched to the same characteristic impedance. Both pads shall be high enough in attenuation value to minimize the error caused by any mismatch of the signal generator and detector. For the majority of measurements, it is recommended that the attenuation of each pad be approximately 10 dB. Tuning stubs may be used in the circuit for impedance matching purposes. Any other method approved by the procuring activity may be used in lieu of that described herein. When the attenuation of the cable under test is less than 1 dB at the test frequency, the attenuation may be measured by the short circuit method.

4.6.7 Velocity (see 3.4.6). The velocity of propagation is determined in terms of the percentage of velocity of wave propagation along the cable to the velocity of an electromagnetic wave in free space. The velocity of propagation in the cable shall be found by resonating a length of cable at a frequency between 10 and 200 MHz with one end short-circuited or open-circuited or by equivalent method subject to the approval of the procuring activity. The same sample may be used for velocity and capacitance measurements.

$$\text{Percent velocity} = \frac{\text{fr x length (ft)}}{2.46\ N}$$

Where:
fr = resonant frequency in MHz.
N = number of quarter wavelengths in the cable.

4.6.8 Capacitance (see 3.4.7). The capacitance of the cable shall be measured to three significant figures, at any one frequency between 1 kilohertz (kHz) and 1 megahertz (MHz) reported in picofarads (pf) per foot. An electrically short piece, that is less than 1/40 of a wavelength of cable, shall be used for this test.

4.6.9 Impedance (see 3.4.8). The characteristic impedance of cable shall be determined by calculation from the capacitance and the velocity propagation measurements using the following formula:

$$Z_o \text{ in ohms} = \frac{101{,}600}{\text{(percent velocity) (capacitance in picofarads per foot (pf/ft)}}$$

Velocity propagation and capacitance values shall be determined as specified in 4.6.7 and 4.6.8.

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4.6.10 VSWR (see 3.4.9). The VSWR of a minimum length of 100 feet of cable shall be measured over the specified frequency range (see 3.1) using a swept frequency technique with capability of measuring a VSWR of 1.04 or less. The measurement system may be in the form of directional coupler(s), hybrid, or reflection coefficient bridge and shall have a directivity of at least 35 dB.

4.6.11 Pressure tightness (when applicable, see 3.5.1). The entire length of finished cable with smooth or corrugated outer conductor shall be pressurized with an internal pressure of 30 psig and the pressure drop shall be measured after a 24-hour period.

4.6.12 Cold bend (when applicable, see 3.5.2). Cables with jackets shall be tested in accordance with 4.6.12.1 or 4.6.12.2, as applicable.

4.6.12.1 Cables (1 5/8 inches nominal diameter or smaller). A section of finished cable, whose length shall be sufficient to make one revolution around a mandrel with a diameter as specified (see 3.1), shall be placed in a chamber and subjected to a temperature of -30 ±2°C for at least 12 hours. After this period, the cable shall be removed and immediately subjected to a 180° bend around the mandrel specified. The cable shall be examined for visual evidence of cracking and breaks in the jacket.

4.6.12.2 Cables (larger than 1 5/8 inches nominal diameter). A section of finished cable, whose length shall be sufficient to make one-half revolution around a mandrel with a diameter as specified (see 3.1), shall be placed in a chamber and subjected to a temperature of -30 ±2°C for at least 12 hours. After this period, the cable shall be removed and immediately subjected at a 180° bend around the mandrel specified. The cable shall be examined for evidence of cracking and breaks in the jacket.

4.6.13 Bending (when applicable, see 3.5.3). Finished cables shall be tested in accordance with 4.6.13.1. When applicable, the alternate procedure, 4.6.13.2 may be used.

4.6.13.1 Procedure. The length of cable shall be sufficient to provide three complete coils around the mandrel specified (see 3.1). One end of the test cable shall be clamped circumferentially at any two points, approximately 45 degrees apart, to the mandrel. The specimen shall then be coiled and uncoiled (the mandrel shall be rotated a minimum of 720 degrees) five times. Although no special tools shall be used during the bending of the cable, a mechanism may be provided to guide the cable on the mandrel. The cable shall then be coiled and uncoiled at a rate between 1 and 5 revolutions per minute (rpm). After the fifth cycle, the sample shall be tested as specified in 4.6.10 and 4.6.12. VSWR may be measured on the sample while coiled.

4.6.13.2 Alternate procedure (for use with cables larger than 1 5/8 inches nominal diameter). The length of cable shall be sufficient to form 180° on the mandrel specified (see 3.1). The test cable shall then be subjected to 10 reverse bends of 180° each on the mandrel. After the fifth cycle, the sample shall be cut open and examined to verify that there is no kinking, wrinkling or cracking.

4.6.14 Temperature cycling (when applicable, see 3.5.4). Cables shall be tested in accordance with 4.6.14.1 or 4.6.14.2, as applicable.

4.6.14.1 Cables 1 5/8 inches nominal diameter and smaller). A length of cable with connectors properly attached, sufficient to make one complete 360 degree turn, shall be coiled on a mandrel of a diameter equal to the diameter specified (see 3.1). The mandrel with cable firmly attached shall be placed in a chamber and subjected to the temperature cycling specified in table VIII. After the cycling has been completed, the cable shall be tested as specified in 4.6.8 and 4.6.10.

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4.6.14.2 Cables (larger than 1 5/8 inches nominal diameter). A length of cable with connectors properly attached, sufficient to make one-half revolution around a mandrel with a diameter as specified (see 3.1), shall be subjected to the temperature cycling specified in table VIII. After the cycling has been completed, the cable shall be tested as specified in 4.6.8 and 4.6.10.

TABLE VIII. Temperature cycling.

Step	-55°C to +80°C	-55°C to +200°C	Time
	(°C)	(°C)	(hours)
1	-55 ±2	-55 ±2	4 - 8
2	25 $^{+10}_{-5}$	25 $^{+10}_{-5}$	4 - 24
3	80 ±2	200 ±5	4 - 8
4	25 $^{+10}_{-5}$	25 $^{+10}_{-5}$	4 - 24

4.6.15 Thermal shock (when applicable, see 3.5.5). A piece of cable, 5 feet long shall be prepared by carefully removing 1 inch of insulation from each end. (For purpose of this test, insulation is defined as all layers of nonconducting material covering the electrical conductor, e. g., primary insulation, all tapes, and braids of the jacket). The specimen shall be formed into a loose coil not less than 1 foot in diameter and shall be laid on a screen for handling throughout the test (see 3.1). A razor blade, or equivalent, held perpendicular to the axis of the specimen shall be used to cut the insulation for the removal operation. The length of exposed conductor at each end of the specimen shall be measured to the nearest 1/8 inch or as specified (see 3.1). The test cable shall be placed for 30 minutes in a preheated air-circulating oven at the temperature specified (see 3.1). It shall then be removed from the oven and within 2 minutes placed in a chamber maintained at -55° ±2°C (-67° ±4°F). The specimen shall be exposed to this temperature for 30 minutes. It shall be removed and stored for a minimum of 30 minutes at room temperature, 20° to 25°C (68° to 77°F). At the conclusion of this cycle, the distance from the end of each layer of insulation to the end of the conductor shall be measured to the nearest 1/8 inch (see 3.1). This thermal shock and the measurements shall be repeated for an additional three cycles (a total of four cycles). Cables shall be examined for flaring of any layer, and checked for measurement variations.

4.6.16 Vibration, high frequency (when applicable, see 3.5.6). Cables shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting of specimen - Secure one end of an appropriate length of cable (supported by three or less rigid clamps on each end) to a vibrating means while the other end is supported by a rigidly stationary surface. A foot of free cable length shall exist between the supporting surfaces. Each end shall describe a 180° nominal bend.
(b) Test-condition letter - B, except 10-55 Hz.
(c) Measurements - Short circuits shall be continually monitored throughout test with a detector applied by a latching relay, conforming to MIL-R-5757/21.

Cables shall be examined for damage.

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4.6.17 Shock (when applicable, see 3.5.7). Cables shall be tested in accordance with method 202 of MIL-STD-202. The following details shall apply:

(a) Mounting method and accessories - Three or less cable straps to provide as rigid support as possible.
(b) Acceleration requirements - 15g.
(c) Number of blows - 18 shocks, 3 in each of 3 mutually perpendicular directions.
(d) Measurements after shock - Cables shall be examined for discontinuity and damage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Packaging, packing and marking. Cable shall be packaged on reels, packed and marked in accordance with the specified level (see 6.2) of protection requirements of MIL-C-12000. In addition cable shall be shipped with a pressure-end-seal on one end and a pressure-end-seal with a testing valve on the other end. The cable shall be pressurized with dry air or suitable gas of at least five psig.

5.1.2 Level C. Cable shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Cables, packaged as specified in 5.1.1, shall be packed in accordance with MIL-C-12000.

5.2.2 Level C. Cable shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This pack shall conform to the applicable carrier rules and regulations.

5.3 Marking (see 6.2). In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-C-12000.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

5.5 Inspection. Inspection of military packaging shall be in accordance with MIL-P-116 (see 4.5.3).

6. NOTES

6.1 Intended use. Cables covered by this specification are intended for use with military communication equipment and include two or more general installation classes: airborne, shipboard, and ground.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number.
(c) Levels of preservation and packaging and packing and applicable marking (see section 5).

6.2.1 Indirect shipment. The preservation, packaging, packing, and marking requirements specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.3 Connectors. Users are cautioned that connectors vary between cable type and the proper connector must be obtained for the cable type actually procured.

6.4 Cross reference. Cross reference information to former classification will be found on the individual specification sheets (see 3.1).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH, EC
Air Force - 11, 17, 85

User activities:
Army - MI, ME, AT, AV
Navy - MC, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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